

PV239 – Using AI in .NET MAUI Development

MAUI Sherpa

- App with useful tools for environment setup and app development
- Doctor - check environment setup (currently available in version 0.7.0)
- Manage Android and iOS related settings - SDKs, keystore, Bundle IDs, certificates, provisioning profiles, secrets, pub
- Send push notifications - Firebase, APNs
- Android SDK packages - check installed tools and versions
- <https://github.com/Redth/MAUI.Sherpa>
- Download from GitHub **Releases**

MAUI Sherpa - Devices

- See connected devices
- Access logs
- Access file system
- Connect to adb shell
- Capture screenshots and screen recordings
- Send deep links
- Manage installed apps

MAUI Sherpa - App Inspector

- Visual tree
- Network
- Profiling
- WebView
- Logs
- Platform information

MAUI DevFlow

- Tool for automating and debugging .NET MAUI apps
- Built to enable AI agents to build, deploy, inspect and debug MAUI apps from terminal
- Used in MAUI Sherpa - App Inspector
- <https://github.com/Redth/MauiDevFlow>

MAUI DevFlow - install

- Install as dotnet tool:

```
dotnet tool install --global Redth.MauiDevFlow.CLI
```

```
dotnet tool install --global androidsdk.tool      # android (SDK, AVD, device management)
```

```
dotnet tool install --global appledev.tools       # apple (simulators, provisioning, certificates)
```

MAUI DevFlow - app setup

- Add nuget package `Redth.MauiDevFlow.Agent`
- Configure MauiProgram.cs

```
#if DEBUG  
builder.AddMauiDevFlowAgent();  
#endif
```

MAUI DevFlow - tools

- maui_screenshot - capture screenshot
- maui_tree - visual tree as structured JSON
- maui_tap - tap a UI element by ID
- maui_fill - fill text into an Entry/Editor/SearchBar
- maui_scroll - scroll a ScrollView (by delta, item index, or position)
- maui_navigate - navigate to a Shell route
- ...
- Complete list: <https://github.com/Redth/MauiDevFlow?tab=readme-ov-file#available-tools>

GitHub Copilot CLI

- One of the clients for GitHub Copilot
- /model - Choose model based on use-case (OpenAI, Anthropic, Google, xAI...)

GitHub Copilot CLI - MCP Servers

- /mcp - add and select active MCP servers
- "Normal" information cap can be about a year old - old .NET versions, old code practices etc.
- Microsoft Learn MCP Server - <https://learn.microsoft.com/en-us/training/support/mcp>
- MAUI DevFlow MCP Server - <https://github.com/Redth/MauiDevFlow?tab=readme-ov-file#mcp-server>

GitHub Copilot CLI - usage

- /plan - create a plan
 - Run first before running the actual work
 - Save the plan file somewhere - to review it and be able to access it in the future
 - Make sure there are verify steps
- autopilot - executes the plan autonomously
- /yolo on - enable all permissions - use at your own discretion

GitHub Copilot CLI - Custom skills

- /skills
 - Create custom skills
 - i.e. how to create a page (create page, VM, register it in Shell...)
 - Don't load too many skills into a single session - context window is small
 - **maui_ai_debugging** skill from MAUI DevFlow - describe what you want to debug and it will do it - clicking, filling texts, navigating etc.

Polypilot

- Multi-agent control plane for GitHub Copilot
- Useful with GIT worktree - separate checkouts for separate branches, multiple tasks running simultaneously locally
- Alternatively use
- Can be accessed from mobile - see current status of sessions, send commands etc.
- <https://github.com/PureWeen/PolyPilot>

VS Code integration

- Add .vscode/mcp.json file

```
{
  "servers": {
    "maui-devflow": {
      "type": "stdio",
      "command": "maui-devflow",
      "args": ["mcp-serve"]
    }
  }
}
```

Punk

- Many of these tools didn't exist at the beginning of this semester
- This all is changing rapidly
- It will accelerate even faster
- Sample code: <https://github.com/jasho/chatr>